List, Dictionary, Tuple, Set

1. Write a program to store 5 student marks in a list and display them.

```
mylist = []
for i in range(1,6):
  marks = int(input(f"Enter marks for student {(i)}: "))
  mylist.append(marks)
print(mylist)
```

```
Enter marks for student 1: 10
Enter marks for student 2: 20
Enter marks for student 3: 30
Enter marks for student 4: 40
Enter marks for student 5: 50
[10, 20, 30, 40, 50]
```

2. Write a program to find the largest number in a list.

```
mylist = [10,44,56,35,35,356,3,5,7,54,754,3,0,56,78]
print("Largest number in list : ", max(mylist))
```

```
Largest number in list :  754
```

3. Write a program to count how many times a particular element appears in a list.

```
mylist = [10,44,56,35,35,356,3,5,7,54,754,3,0,56,78]
print("appearance of en element in list : ", mylist.count(35))
```

```
appearance of en element in list :  2
```

4. Write a program to add a new element to a list and remove an existing element.

```
mylist = ['10','44','56','35']
print(mylist)
ele = input("Enter element to be add : ")
mylist.append(ele)
print(mylist)
rep = input("Enter element to remove : ")
mylist.remove(rep)
print(mylist)
```

```
['10', '44', '56', '35']
Enter element to be add : 20
['10', '44', '56', '35', '20']
Enter element to remove : 10
['44', '56', '35', '20']
```

5. Write a program to calculate the sum of all elements in a list.

```
mylist = [10,44,56,35,35]
print("Sum of all elements : ", sum(mylist))
```

```
Sum of all elements :  180
```

6. Write a program to create a tuple of 5 numbers and display the second element.

```
mytuple = ("apple", "banana", "cherry", "dragon fruit", "Papaye")
print(mytuple[1])
```

```
banana
```

7. Write a program to check whether an element exists in a tuple.

```
mytuple = ("apple", "banana", "cherry", "dragon fruit", "Papaye")
ele = input("Enter element : ")
if (ele in mytuple):
  print("Element exist in tuple!")
else:
  print("Element does not exist in tuple!")
```

```
Enter element : apple
Element exist in tuple!
```

8. Write a program to convert a tuple into a list.

```
mytuple = ("apple", "banana", "cherry", "dragon fruit", "Papaye")
print("Tuple :", mytuple)
mylist = list(mytuple)
print("List :", mylist)
```

```
Tuple : ('apple', 'banana', 'cherry', 'dragon fruit', 'Papaye')
List : ['apple', 'banana', 'cherry', 'dragon fruit', 'Papaye']
```

9. Write a program to find the length of a tuple.

```
mytuple = ("apple", "banana", "cherry", "dragon fruit", "Papaye")
print("Length of tuple :", len(mytuple))
```

```
Length of tuple : 5
```

10. Write a program to explain why tuple is immutable with an example.

```
tuple1 = (10,20,30)
print(tuple1)
tuple1[1]=40
print("Updated",tuple1)
```

```
(10, 20, 30)
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipython-input-4061893314.py in <cell line: 0>()
      1 tuple1 = (10,20,30)
      2 print(tuple1)
----> 3 tuple1[1]=40
      4 print("Updated",tuple1)

TypeError: 'tuple' object does not support item assignment
```

11. Write a program to store student details using a dictionary and display them.

```
myDict = {
    "Alice": 30,
    "Bob": 40
}
name = input("Enter a name : ")
marks = int(input(f"enter marks for {name} : "))
myDict[name] = marks
print(myDict)
```

```
Enter a name : john
enter marks for john : 30
{'Alice': 30, 'Bob': 40, 'john': 30}
```

12. Write a program to add a new key-value pair in a dictionary.

```
myDict = {
    "Alice": 30,
    "Bob": 40
}
name = input("Enter a name : ")
marks = int(input(f"enter marks for {name} : "))
myDict[name] = marks
print(myDict)
```

```
Enter a name : john
enter marks for john : 50
{'Alice': 30, 'Bob': 40, 'john': 50}
```

13. Write a program to update the value of an existing key in a dictionary.

```
myDict = {
    "Alice": 30,
    "Bob": 40
}
print(myDict)
```

```
name = input("Enter a name to be updated : ")
marks = int(input(f"enter new marks for {name} : "))
myDict[name] = marks
print(myDict)
```

```
{'Alice': 30, 'Bob': 40}
Enter a name to be updated : Bob
enter new marks for Bob : 50
{'Alice': 30, 'Bob': 50}
```

14. Write a program to check whether a key exists in a dictionary.

```
myDict = {
    "Alice": 30,
    "Bob": 40
}
key = input("Enter a key to be find : ")
if key in myDict:
  print("Key Found!")
else:
  print("Key not found!")
```

```
Enter a key to be find : Bob
Key Found!
```

15. Write a program to print all keys and values of a dictionary.

```
myDict = {"alice": 1 , "Bob": 2, "Charlie": 3}
for key, value in myDict.items():
  print(f"Key : {key},  Value : {value}")
```

```
Key : alice,  Value : 1
Key : Bob,  Value : 2
Key : Charlie,  Value : 3
```

16. Write a program to create a set with duplicate values and display the output.

```
myset = {10,20,10,3}
print(myset)
```

```
{10, 3, 20}
```

17. Write a program to add a new element to a set.

```
myset = {10,20,10,3}
ele = int(input("Enter element to be add : "))
myset.add(ele)
print(myset)
```

```
Enter element to be add : 40
{40, 10, 3, 20}
```

18. Write a program to remove an element from a set.

```
myset = {10,20,10,3}
ele = int(input("Enter element to be remove : "))
myset.remove(ele)
print(myset)
```

```
Enter element to be remove : 20
{10, 3}
```

19. Write a program to find the union of two sets.

```
myset1 = {10,20,10,3}
myset2 = {"aa", "b", "cc"}
set3 = myset1.union(myset2)
print(set3)
```

```
{'cc', 3, 20, 'b', 10, 'aa'}
```

20. Write a program to find the intersection of two sets.

```
myset1 = {10,20,10,3}
myset2 = {10,40,30}
set3 = myset1.intersection(myset2)
print(set3)
```

```
{10}
```